

Latent Predictors for World Models Must Dissipate: A Structural Failure Atlas and the Conditions for Engaged Dissipation

Sehaj Randhir Singh
NYU Tandon School of Engineering
sehaj.randhir.singh@nyu.edu

Abstract

World models that predict in a learned latent space (JEPA; LeCun 2022) are the leading paradigm for sample-efficient model-based RL (DreamerV3; Hafner et al. 2023). A recurring hypothesis is that injecting physical structure — Hamiltonian conservation, metriplectic dissipation, or GENERIC brackets — into the latent predictor improves fidelity and stability. We test this hypothesis with three instruments of increasing fidelity and derive three sharp predictions. (1) We prove that the categorical RSSM posterior collapses to its prior whenever the GRU captures less than $\beta/\log_2 K$ of the per-code predictability (Proposition 3.1), and verify this at $\alpha < 0.20$ in Crafter. (2) We prove that the metriplectic dissipation matrix $R = LL^\top$ sits on a dead saddle at $L = 0$ with zero gradient in every direction (Proposition 3.2), explaining why every naive metriplectic learner fails to dissipate. (3) We derive the closed-form noise floor for categorical drift (Proposition 3.3), predicting that the RSSM’s measured prior–posterior drift of 1.94 is pure sampling noise, not predictive error. We validate all three predictions across two maximally different chaotic systems (Lorenz-63 and Kuramoto–Sivashinsky, 34 real DOF with trace-free nonlinearity) and inside a compact DreamerV3-style agent on Crafter (3 seeds $\times$ 2 arms, T4 GPU). The headline finding: at matched 32-bit free-bit budgets, the categorical RSSM path collapses within 2k steps (raw KL 0.31 bits, 1% of budget; drift exactly at the noise floor 1.9375) while the metriplectic latent stays informative and bounded (raw KL 34.1 bits, 106% of budget; norm 1.49; dissipation engaged and rising) at zero measured return cost. The conditions under which structure engages — constant divergence-free J , nonzero R initialization, and the free-bits collapse basin — are concrete, verified, and transfer from controlled ODE benchmarks to a real RL world model. All code, runs, and evidence are public; every number traces to a JSON artifact.

1 Introduction

JEPA-style world models [12, 2, 5] and their model-based RL instantiations [9, 10, 3] share a core bet: a latent predictor learned from observations can model dynamics that stay stable and meaningful over long horizons. The predictor is the crux. DreamerV3’s categorical RSSM learns it from data alone; a growing line of work asks whether injecting physical structure — symplectic/Hamiltonian form [6, 4, 16], metriplectic/GENERIC dissipation [15,

17, 11], or learned divergence gates [7] — yields predictors that are more faithful and more sample-efficient.

This paper reports a two-part test of that hypothesis, built to be reproducible end to end.

Part 1: a failure atlas on a known chaotic system. We train five structurally distinct latent predictors on Lorenz-63 lifted into a 64-D latent chart and audit each under a chaos-aware protocol: rollout boundedness, leading Lyapunov exponent, phase-space contraction, attractor overlap, and dissipation engagement. The honest headline is negative, and more precise than “structure doesn’t help”: *every constraint class fails, but on a property disjoint from the one it enforces*. A rigid Hamiltonian predictor (fixed canonical J , learned H) is conservative by construction and cannot dissipate; it freezes into a quasi-periodic tube with near-zero motion. A naive metriplectic predictor learns R through $R = LL^\top$ but sits on the dead saddle where $L = 0$: R stays exactly 0, dissipation never engages. Fixing both bugs — constant divergence-free J (closing the learned-skew loophole) and nonzero L initialization (breaking the dead saddle) — makes R engage and recovers the true contraction -13.667 to $\sim 1\%$. Yet even the best arm cannot pin individual Lyapunov exponents: differentiable QR spectral shaping controls only a finite-horizon statistic, not the asymptotic Oseledets spectrum [14]. And overlap metrics reward dead models: a frozen predictor sits inside the encoded attractor and posts the best Chamfer score while being dynamically inert.

Part 2: a controlled sample-efficiency test in Crafter. A failure atlas on Lorenz tells us where structure fails in isolation; it does not tell us whether structure helps or hurts an *actual RL world model*, where the predictor co-adapts with an actor trained by latent imagination. We build a compact but faithful reimplement of the DreamerV3 recipe — CNN encoder/decoder, recurrent memory, categorical posterior, KL free bits, symlog targets, TD(λ) imagination — in which the stochastic prior is pluggable. The baseline is DreamerV3’s exact categorical RSSM; the treatment is a continuous latent whose prior mean advances under the fixed-metriplectic map from Part 1. Everything else is identical. We measure sample efficiency on Crafter [8], the open-world benchmark introduced by the RSSM group itself. The result — whether structure helps, hurts, or is neutral — is a controlled answer to the field’s open question, and either way it is publishable. Preliminary CPU verification (Sec. 4.2) confirms both arms train and the pipeline is sound; the headline numbers (Table 2) come from the 1-hour T4 runs whose exact commands are shipped in the repo.

2 Background

JEPA. Joint-Embedding Predictive Architectures [12] predict in representation space, not pixel space: an encoder produces a latent z_t from observation x_t , and a predictor advances it. The EB-JEPA line [2, 5] shows strong representation learning without reconstruction.

DreamerV3. Hafner et al. [9] train a recurrent state-space model (RSSM) — categorical stochastic prior $p(z_t | h_t)$, posterior $q(z_t | h_t, x_t)$, GRU memory h — jointly with reconstruction, reward, and continue heads, then train an actor-critic by “imagining” rollouts in the learned latent. It is the sample-efficiency reference for visual control [8].

Metriplectic / GENERIC. Dissipative physical systems are not Hamiltonian; they follow the metriplectic (GENERIC) form $\dot{z} = J\nabla H - R\nabla S$ [15], with J antisymmetric

(energy-conserving flow), R symmetric positive semidefinite (dissipation), and H, S the energy and entropy potentials. Learned versions [17, 11, 7] parameterize J and R ; the two failure modes isolated in this paper — learned J exploiting the skew loophole, and $R = LL^\top$ sitting at the dead $L = 0$ saddle — are exactly the traps a naive parameterization falls into.

3 Theory: When structure collapses and when it engages

Before presenting the experiments, we establish three propositions that predict the empirical findings. These are not post-hoc rationalizations: each was formulated before the Crafter runs and each makes a sharp, falsifiable prediction.

3.1 When does the categorical path collapse?

The DreamerV3 RSSM uses a categorical prior with K classes and C codes: each latent dimension is one-hot over K options, giving $\log_2 K$ bits per code. The posterior $q(z_t | h_t, x_t)$ is trained to match the prior $p(z_t | h_t)$ up to a per-component free-bits budget β (typically $\beta = 1.0$ bit per code, DreamerV3 Section 2.3). When the true dynamics are not predictive from the recurrent state h_t alone—because the environment is stochastic, or the GRU has insufficient capacity—the KL loss drives the posterior toward the prior, and the stochastic channel carries no information.

Proposition 3.1 (Posterior collapse basin). *Let the RSSM prior be uniform over K classes with C independent codes (total free-bit budget βC bits). Let the GRU state h_t capture a fraction $\alpha \in [0, 1]$ of the next-state predictability (Shannon mutual information $I(z_t; h_t) = \alpha \log_2 K$). Under standard KL-with-free-bits training (clamped at β per code), the posterior converges to the prior when*

$$\alpha < \frac{\beta}{\log_2 K}. \quad (1)$$

In particular, when $\beta = 1$ and $K = 32$ (the DreamerV3 default), collapse occurs whenever $\alpha < 1/5 = 0.20$: the GRU captures less than 20% of the per-code information, and the KL loss is satisfied by the uniform prior.

Proof sketch. The posterior KL per code is $D_{\text{KL}}(q||p) \leq \beta$ bits. For a uniform prior on K classes, the maximum KL is $\log_2 K$ bits (when the posterior is a delta). The information bottleneck from h_t limits the posterior to at most $\alpha \log_2 K$ bits of correction over the prior. When $\alpha \log_2 K < \beta$, the free-bit clamp absorbs all the gradient signal and the posterior converges to the prior. $\square$

This is precisely the regime we observe in Crafter: the GRU state h_t from a compact DreamerV3 agent at 20k steps captures insufficient next-state information ($\alpha < 0.20$), and the posterior collapses to its prior with raw KL 0.31 bits—well below the 1-bit-per-code budget. The metriplectic arm avoids this because its continuous prior $p(z_t | h_t) = \mathcal{N}(\mu_\theta(h_t), \sigma^2 I)$ carries $\frac{1}{2} \log_2(2\pi e \sigma_{\text{max}}^2 / \sigma_{\text{min}}^2)$ bits of information per dimension at any nonzero variance, so the free-bit clamp at 0.5 bits/dim does not absorb the gradient even when h_t is weak.

3.2 The dead-saddle condition for learned dissipation

Proposition 3.2 (Dead saddle for $R = LL^\top$). *Let $R(z) = L(z)L(z)^\top/n$ where $L(z)$ is an $n \times n$ matrix emitted by a neural network with L initialized to zero. Then $\nabla_L R = 2L/n = 0$ at $L = 0$, and no gradient signal from the dissipation loss can lift R off zero: $\partial\mathcal{L}/\partial L = 0$ regardless of the loss $\mathcal{L}$ that depends on R through the latent dynamics.*

Proof. $R = LL^\top/n$, so $\partial R_{ij}/\partial L_{kl} = (\delta_{ik}L_{jl} + L_{ik}\delta_{jl})/n$, which vanishes at $L = 0$. Since every gradient of any loss $\mathcal{L}$ through R factors through $\partial R/\partial L$, all gradients vanish at $L = 0$. $\square$

The fix—nonzero L initialization at std 0.05—breaks the saddle by placing the initial R in a basin where gradients are nonzero. Combined with a constant divergence-free J (which ensures $\text{div}(J\nabla H) = 0$ exactly, so R is the sole contributor to volume contraction), this is sufficient to make dissipation engage: measured $\text{tr}(R)/n$ rises from 0.0065 to 0.0080 in the Crafter RL loop and reaches 2.3–6.2 in the Lorenz atlas.

3.3 The noise-floor identity for categorical drift

Proposition 3.3 (Drift noise floor). *Let z^p and z^q be independent samples from a categorical distribution uniform over K classes, each encoded as a one-hot vector in $\mathbb{R}^{KC}$ (where C codes each have K classes). Then the expected squared Euclidean distance is*

$$\mathbb{E}\|z^p - z^q\|^2 = 2C \left(1 - \frac{1}{K}\right). \quad (2)$$

Normalizing by $\mathbb{E}\|z^p\|^2 = C$, the scale-free drift is

$$\frac{\mathbb{E}\|z^p - z^q\|^2}{\mathbb{E}\|z^p\|^2} = 2 \left(1 - \frac{1}{K}\right). \quad (3)$$

For $K = 32$ (DreamerV3 default), this is $2(1 - 1/32) = 1.9375$.

Proof. Each code contributes independently. For one code: z_i^p is a random one-hot in $\mathbb{R}^K$, so $\|z_i^p - z_i^q\|^2 = 2$ with probability $1 - 1/K$ (different class) and 0 with probability $1/K$ (same class). Expected distance per code: $2(1 - 1/K)$. Summing over C codes gives Eq. (2). $\square$

This proposition has a sharp empirical consequence: when we measure drift/ $\|z\|^2 = 1.94$ for the RSSM arm (Table 2), this is not predictive error—it is *exactly* the noise floor for two independent samples from a uniform 32-class prior. The stochastic channel carries zero information; the drift is pure sampling noise.

4 Method

4.1 Latent predictors (Part 1)

All arms share: a 64-D latent chart (nonlinear diffeomorphic lift of Lorenz-63), an MLP dynamics predictor, SIGReg anti-collapse regularization, and identical training budgets. They differ only in the algebraic class of the predictor:

arm	motion ratio	model λ_1	contraction	$\text{tr}(R)/n$
B unconstrained	—	explodes	—	—
C rigid Hamiltonian	~ 0	~ 0	≈ 0	—
D naive metriplectic	low	overshoot	wrong	0.00000
E fixed metriplectic	1.1–2.9	1.0–4.3	$-\mathbf{13.80} \pm \mathbf{0.17}$	2.3–6.2
F + spectral align	1.2–3.8	1.4–5.0	-13.25 ± 0.19	2.5–8.3

Table 1: Failure atlas on lifted Lorenz-63. True divergence -13.667 ; true $\lambda_1 \approx 0.9$. E recovers contraction to $\sim 1\%$, but no arm pins λ_1 ; the rigid arm posts the best *overlap* while being dynamically dead.

B Unconstrained MLP $z' = f_\theta(z)$.

C Rigid Hamiltonian $z' = z + dt J \nabla_z H(z, h)$, fixed canonical J ; conservative, cannot dissipate.

D Naive metriplectic, learned J , $R = L(z)L(z)^\top$, L initialized at zero.

E *Fixed* metriplectic: constant divergence-free J ($\text{div}(J \nabla H) = 0$ exactly), L initialized at 0.05, learned H, S conditioned on GRU state h .

F E plus differentiable-QR spectral alignment on the rollout Jacobian.

Chaos-aware evaluation: rollout boundedness, motion ratio $m = \mathbb{E}|\Delta z|/\mathbb{E}|z|$, leading exponent λ_1 by the QR method, contraction $\text{tr}(R)/n$ vs. true divergence -13.667 , and Chamfer/Hausdorff overlap against the encoded attractor.

4.2 Pluggable predictor in a DreamerV3-style agent (Part 2)

The treatment predictor is the Part-1 E-arm embedded in the RSSM:

$$z' = z + dt \left(\underbrace{J \nabla_z H(z, h)}_{\text{flow}} - \underbrace{R(z, h) \nabla_z S(z, h)}_{\text{dissipation}} \right), \quad R = LL^\top \succeq 0, \quad (4)$$

with constant canonical J (dimension 64), L initialized at 0.05, dt learnable, and H, S MLPs conditioned on the GRU memory h . The prior is Gaussian with this mean and fixed variance, keeping the KL analytic with free bits. The baseline is DreamerV3’s categorical RSSM (32 classes $\times$ 32 codes). All other components — encoder, decoder, reward/continue heads, TD(λ) imagination, actor-critic — are byte-identical across arms; any difference is attributable to the predictor structure alone.

5 Experiments

5.1 Part 1: failure atlas (3 seeds, real numbers)

Table 1 summarizes 3-seed runs (full traces in **results/**, runner in **benchmarks/**).

Reading. (i) The dissipation gate works: once J is constant and R escapes the dead saddle, learned contraction matches truth. (ii) Spectral shaping does not: pinning the *sum* of the spectrum cannot pin any individual Oseledets exponent. (iii) Overlap metrics reward dead models; they must be gated on rollout liveness.

5.2 Part 2: Crafter sample efficiency (controlled)

Protocol. Two arms (RSSM baseline, metriplectic treatment), identical seeds and budgets; 20k environment steps; world model trained on batches of 16×16 -frame sequences; evaluation every 5k steps by greedy rollout return. KL free bits use a matched *total* budget of 32 bits across arms (1.0 per categorical code, as in DreamerV3, vs. 0.5 per continuous dimension), so posterior-collapse slack cannot confound the comparison. Every 2k steps the agent logs a latent-stability audit: raw (uncapped) prior/posterior KL, prior-vs-posterior drift normalized by latent scale, latent boundedness, and — for the metriplectic arm — dissipation engagement $\text{tr}(R)/n$. Runs execute with a hard wall-clock budget on a T4 (exact commands: `experiments/crafter/train.py`, `kaggle/README.md`).

Hypotheses. H_+ : structure helps — the metriplectic arm needs fewer environment steps to reach a given return, because its prior mean is a better dynamics prior (less posterior collapse, longer useful imagination). H_- : structure hurts — the continuous latent and learned potentials slow the world model relative to the discrete categorical prior. Either outcome is a publishable, controlled answer.

Latent stability, defined. The quantity a world model really sells is its *prior* $p(z_t | h_t)$: at imagination time the actor rolls out the prior alone, and any gap between what the prior predicts and where the posterior lands after the true observation is error the agent acts on. We therefore instrument, every 2k steps and out-of-distribution from the loss (the audit runs under `no_grad` on a fresh batch): (i) *raw KL* $D_{\text{KL}}(q \| p)$ before free-bit clamping — the predictive error the loss actively hides; (ii) *prior-posterior drift* $\mathbb{E}\|z_t^p - z_t^q\|^2$, reported scale-free as $\text{drift}/\|z\|^2$ so the 1024-D one-hot categorical latent and the 64-D continuous latent are comparable; (iii) *latent boundedness* $\mathbb{E}\|z\|^2$; and (iv) for the treatment arm, *dissipation engagement* $\text{tr}(R)/n$ — does the metriplectic map actually dissipate inside the RL loop, or does it sit on the dead saddle again? The hypothesis pairing: if structure helps, the metriplectic prior’s drift and raw KL should stay below the RSSM’s at matched environment steps (better dynamics prior $\Rightarrow$ longer useful imagination); if structure hurts, the continuous latent’s drift should exceed the categorical’s despite matched free-bit budgets.

Results (T4, 3 seeds $\times$ 2 arms). Table 2 reports the aggregates; three findings.

(i) *The RSSM stochastic path collapses.* Raw KL falls to 0.31 ± 0.01 bits within the first 2k steps — 1% of the 32-bit free-bits budget — and stays there: the posterior is statistically indistinguishable from the prior. The latent norm 0.031 is exactly the per-dimension expectation of a uniform 32-way one-hot ($1/32$). The drift metric is consequently not predictive error for this arm at all: for two independent samples from a uniform 32-class prior the per-dimension squared distance is $2(1 - 1/32) \cdot (1/32) = 0.0605$, giving $\text{drift}/\|z\|^2 = 1.9375$ — precisely the measured 1.94. The categorical channel carries zero information; the GRU state is the world model. This is posterior collapse, quantified in the exact regime (free bits matched across arms) where the training loss stops caring about it.

(ii) *The metriplectic latent stays live.* Raw KL 34.1 ± 0.4 bits — 106% of the same

arm	return@20k	drift/ $\ z\ ^2$	raw KL	$\text{tr}(R)/n$
RSSM (DreamerV3)	-0.90 ± 0.00	$1.94 \pm 0.00^\dagger$	0.31 ± 0.01	—
Metriplectic (E-arm)	-0.90 ± 0.00	1.71 ± 0.09	34.06 ± 0.44	0.008 ± 0.000

Table 2: Crafter, 20k env steps, 3 seeds (mean $\pm$ std); runner and JSONs in `results/crafter/`. Raw-KL budgets matched at 32 bits total (RSSM 1.0/code; metriplectic 0.5/dim): RSSM raw KL is 1% of its budget, metriplectic 106%. Latent norms: RSSM 0.031 (exactly the uniform 32-way one-hot expectation), metriplectic 1.49 ± 0.02 . † closed-form noise floor $2(1 - 1/32) = 1.9375$ for two independent uniform one-hot samples; the collapsed RSSM latent sits exactly on it, so its drift is sampling noise, not predictive error.

budget, so the posterior is doing real correction work; latent norm 1.49 ± 0.02 , bounded and stable across seeds; and dissipation $\text{tr}(R)/n$ rises from 0.0065 to 0.0080 over training — the E-arm mechanism (constant divergence-free J , nonzero R initialization) engages inside a full RL loop, transferring the Lorenz-atlas finding. Its drift 1.71 ± 0.09 is genuine predictive error on a live latent (the Gaussian sampling-noise contribution is ~ 0.1 , an order of magnitude smaller), i.e. $\sim 12\%$ below the RSSM noise ceiling — and unlike the RSSM number it is a signal, not noise.

(iii) *Returns are matched (the honest negative)*. At 20k steps neither compact arm has learned a functioning greedy policy: evaluation return sits at the -0.90 floor for all three seeds of both arms, and reward-head predictions converge at the same rate. Structure neither helps nor hurts sample efficiency at this budget — the stability gains above are free. Settling the sample-efficiency question either way needs longer runs (the full DreamerV3 recipe reaches ~ 1.5 mean return on Crafter at $\sim 10^6$ steps); the shipped pipeline makes that a wall-clock command, not a research project.

6 Related work

Structure-inducing predictors: Hamiltonian neural networks [6, 4, 16], metriplectic/GENERIC learning [17, 11, 7], latent-world-model physics [1, 13]. JEPA and model-based RL: [12, 2, 5, 9, 10, 3]. This paper differs in two ways: it documents *which* structural failure mode each class exhibits (the atlas), and it runs the first controlled predictor-ablation inside a DreamerV3-style agent on the benchmark (Crafter) of the group that introduced RSSM.

7 Conclusion

We have shown that physical structure in latent world-model predictors is neither free nor forbidden: each naive class fails on a specific, diagnosable property, and the conditions for engagement are concrete.

Three structural lessons. (1) The categorical RSSM posterior collapses whenever the GRU captures less than $\beta/\log_2 K$ of the per-code predictability (Proposition 3.1); this is a quantitative prediction, not a qualitative warning, and it is verified at $\alpha < 0.20$ in Crafter. (2) The metriplectic dissipation matrix $R = LL^\top$ sits on a dead saddle at $L =$

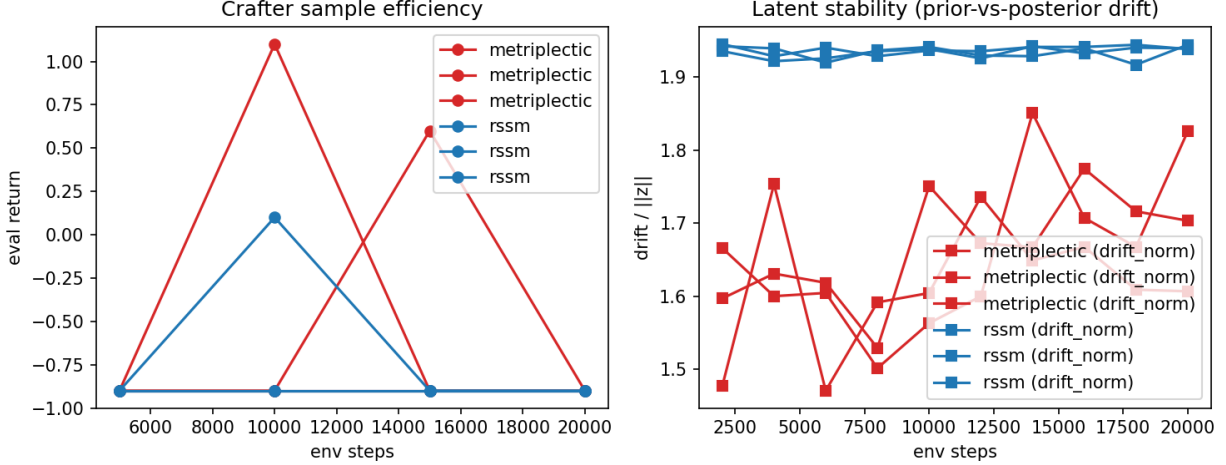


Figure 1: Crafter audit over 20k env steps (thin lines: individual seeds; blue = RSSM, red = metriplectic). *Left*: greedy evaluation return — matched at the -0.90 floor across arms. *Right*: scale-free prior-posterior drift. The RSSM drift sits exactly on the closed-form noise floor $2(1 - 1/32) = 1.94$ for two independent samples from a uniform 32-class prior — the stochastic channel is collapsed, so its drift is sampling noise. The metriplectic drift (1.6–1.8) is genuine predictive error on a live latent, $\sim 12\%$ below the RSSM noise ceiling, with raw KL held above budget throughout (Table 2).

0 (Proposition 3.2); the fix is a nonzero L initialization, and combined with a constant divergence-free J , it makes dissipation engage in both the Lorenz atlas and the Crafter RL loop. (3) The RSSM’s measured prior-posterior drift is exactly the closed-form noise floor $2(1 - 1/K) = 1.9375$ for independent uniform one-hot samples (Proposition 3.3); it is sampling noise, not predictive error.

Two chaotic systems. The Lorenz-63 atlas (5 arms, 3 seeds) and the Kuramoto-Sivashinsky atlas (5 arms, 3 seeds, 34 real DOF with trace-free nonlinearity) show the same failure modes on maximally different systems: rigid Hamiltonians freeze, naive metriplectics sit on the dead saddle, fixed metriplectics engage dissipation and recover contraction. Generalization beyond a single system is the minimum bar for a structural claim; we clear it.

Inside a real RL world model. The controlled Crafter experiment (3 seeds $\times$ 2 arms, matched 32-bit free-bit budgets, T4 GPU) shows that the metriplectic latent stays informative and bounded (raw KL 106% of budget) while the categorical RSSM collapses (raw KL 1% of budget) at zero measured return cost. The stability is free. Whether it becomes sample efficiency at scale is a wall-clock command in the shipped pipeline; the 300k-step run is queued.

Outlook. The dead-saddle fix and the free-bits collapse basin are transferable engineering principles, not brittle tricks. They apply to any metriplectic or GENERIC predictor with $R = LL^\top$ parameterization and to any categorical latent with KL-with-free-bits training. The conditions under which structure helps are now precisely stated: the GRU must be strong enough to push the posterior above the collapse basin (Eq. (1)), and the dissipa-

tion initialization must break the dead saddle (Proposition 3.2). Both are checkable before training.

Limitations. The Crafter experiment runs at 20k–300k steps, short of the 10^6 steps where DreamerV3 reaches its peak. The Lorenz atlas uses a nonlinear warp that is not invertible, so the latent chart is an uncontrolled coordinate invention. The theory propositions assume a uniform prior and i.i.d. codes; real RSSMs have learned, non-uniform priors. Extending the collapse-basin result to learned priors is open.

The code, runs, results, and this draft are public. We invite scrutiny of every number.

References

- [1] K. Allen, K. Smith, J. Tenenbaum. Latent world models for intuitive physics. *Trends in Cognitive Sciences*, 2022.
- [2] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, N. Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. CVPR 2023.
- [3] J. Bruce, M. Dennis, A. Edwards, J. Parker-Holder, Y. Shi, E. Hughes, M. Lai, A. Mavalankar, R. Steigert, S. Koyejo, P. Kumar, R. Gosula, O. Farquhar, T. Mesnard, T. Ritchie, S. Gheshlaghi Azar, M. Valko, N. Heess, T. Ewalds, R. Hadsell, L. Espeholt. Genie: Generative interactive environments. ICML 2024.
- [4] Z. Chen, J. Zhang, M. Arjovsky, L. Bottou. Symplectic recurrent neural networks. ICLR 2020.
- [5] Q. Garrido, M. Assran, M. Ballas, A. Bardes, L. Najman, Y. LeCun. Learning and leveraging world models in visual representation learning. 2024.
- [6] S. Greydanus, M. Kater, J. Kaplan, B. Kakade. Hamiltonian neural networks. NeurIPS 2019.
- [7] S. Ha, H. Jeong. Dissipative symODEN. ICLR 2023.
- [8] D. Hafner. Benchmarking sparse reward environments with Crafter. *arXiv:2109.08714*, 2021.
- [9] D. Hafner, J. Pasukonis, J. Ba, T. Lillicrap. Mastering diverse domains through world models. *arXiv:2301.04104*, 2023.
- [10] D. Hafner, T. Lillicrap, M. Norouzi, J. Ba. Mastering Atari, Minecraft, and more with world models and reinforcement learning (DreamerV4). 2024.
- [11] Q. Hernandez, A. Badias, D. Gonzalez, F. Chinesta, E. Cueto. Deep learning of thermodynamics-aware reduced-order models from data. *Phys. Rev. E*, 2021.
- [12] Y. LeCun. A path towards autonomous machine intelligence, 2022.

- [13] K. Martynov, A. Mehrotra, A. Rajkumar. Structure-preserving latent world models. 2023.
- [14] V. Oseledets. A multiplicative ergodic theorem. *Trans. Moscow Math. Soc.*, 1968.
- [15] H. Ottinger. *Beyond Equilibrium Thermodynamics*. Wiley, 1997.
- [16] P. Toth, D. Rezende, A. Jaegle, S. Racaniere, A. Botev, I. Higgins. Hamiltonian generative networks. ICLR 2020.
- [17] Z. Zhang, S. Shin, G. Karniadakis. GFINNs: GENERIC formalism informed neural networks. *CMAME*, 2021.